Ciências ULisboa
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Student Academic Success Knowledge Graph

Modeling the Interplay of Socioeconomic, Behavioral, and Educational Factors

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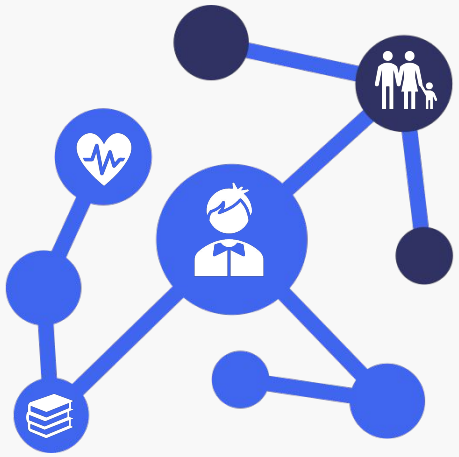
Motivation

Motivation

Academic success is not determined by a single factor. It's the result of a complex interplay between socioeconomic background, behavioral patterns, health, family support, and institutional context.

Traditional machine learning models tend to analyze these features independently, often overlooking the relationships among them.

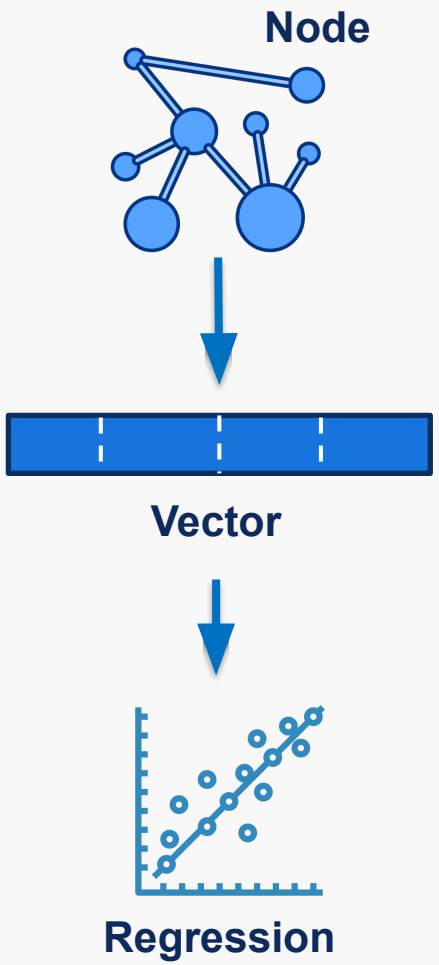
Our goal is to capture these interdependencies using a knowledge graph, which models entities and their connections semantically.



Goals

GOALS

1. Design a **domain-specific ontology** capturing the educational ecosystem.
2. Build a knowledge graph from **real-world data**, representing students and their contexts.
3. Use **embedding techniques** to convert the graph structure into vectors.
4. Apply **regression models** to predict final grades.
5. Extract insights into which factors most influence **student outcomes**.



Dataset

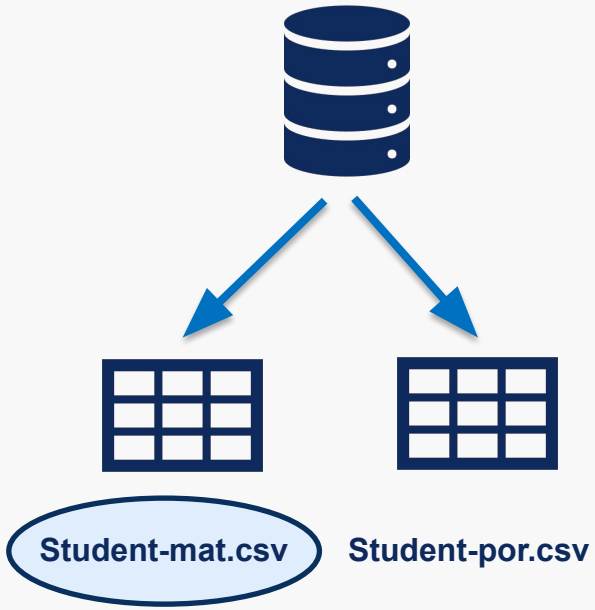
DATASET

We used the Student Alcohol Consumption dataset from Kaggle, which contains data on **395 students** enrolled in a math course.

Attributes include:

- **Personal:** age, gender, absences
- **Family:** parental status, education, relationship
- **Behavioral:** alcohol consumption, social activities
- **Academic:** grades G1, G2, G3
- …

We chose **G3**, the final grade, as the target variable to predict.



Ontology

ONTOLOGY

We created an ontology first with **Protégé** but later switched to Python's **rdflib**.

Key concepts include:

- **Classes:** Student, *Parent*, *School*, *StudyHabit*, *HealthStatus*, etc.
- **Object properties:** *attends*, *hasGuardian*, *hasJob*, etc.
- **Datatype properties:** *hasAge*, *hasAlcoholUse*, hasInternet etc.

We prioritized **clarity**, **modularity**, and **reusability**, ensuring the model could be expanded and reasoned over.

The **ontology** is the semantic backbone of our knowledge graph.

KG Construction

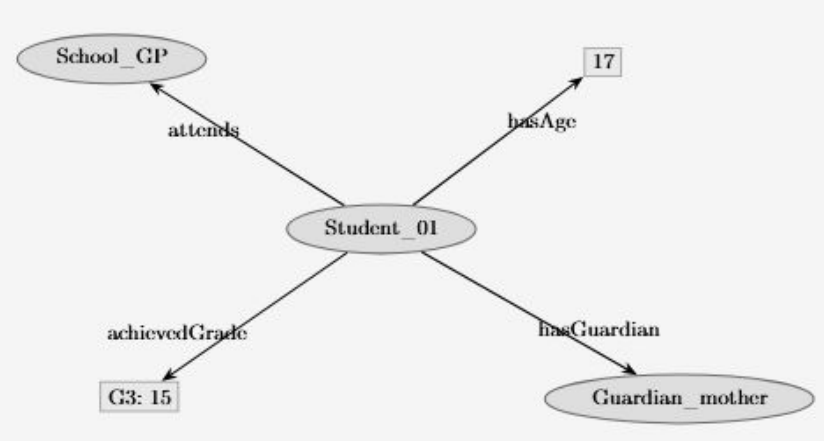
KG CONSTRUCTION

Each student becomes a central node in the knowledge graph, linked to their **school**, **guardians**, **behaviors**, and **grades**.

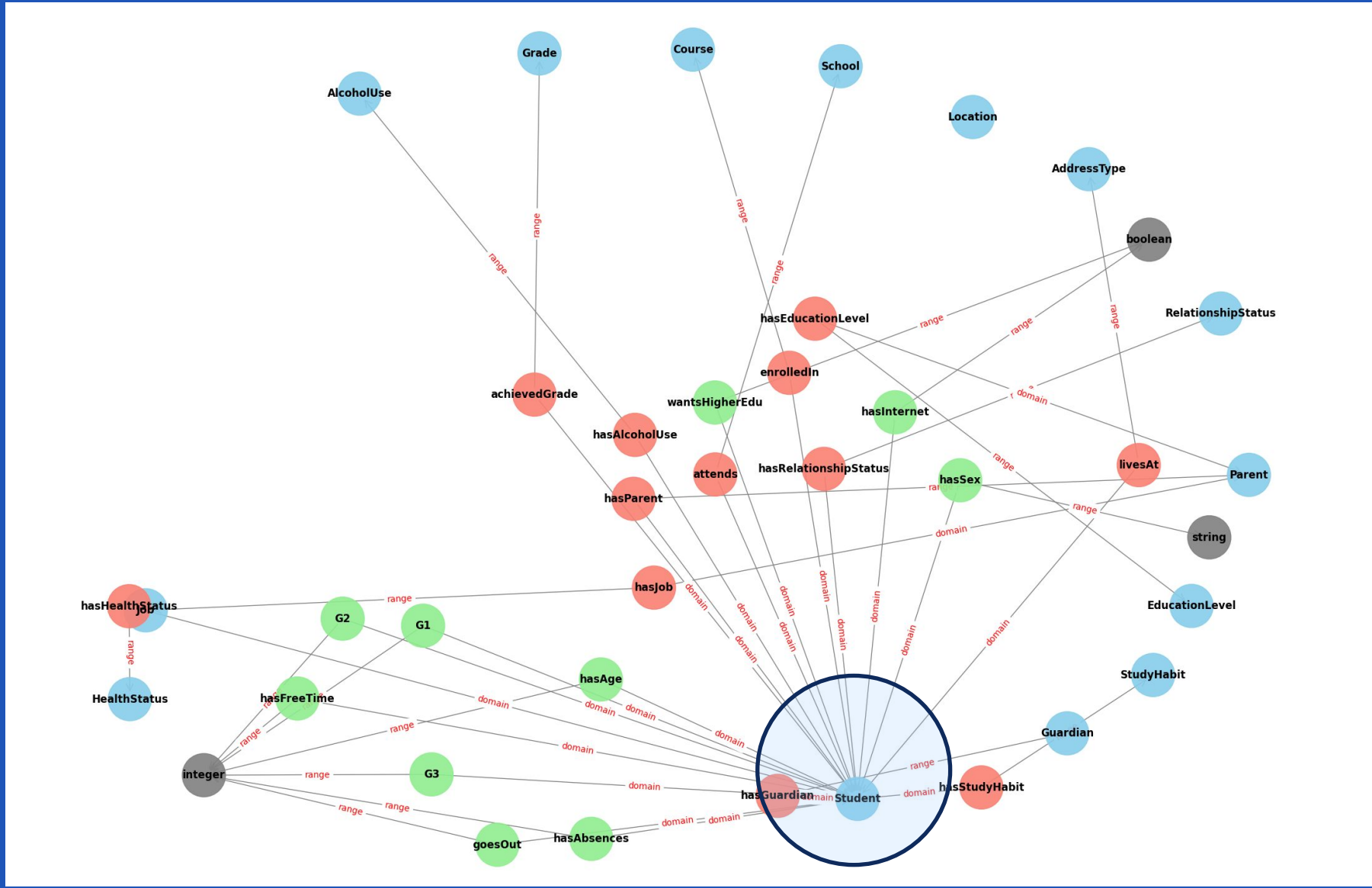
We converted CSV data directly into RDF triples:

- **Object triples:** Student_01→ attends→ School_GP, → achievedGrade→ G3_01
- **Datatype triples:** Student_01 →hasAge→ 17, G3_01→ hasGradeValue→ 15

This rich structure preserves relational context and allows logical reasoning and querying using **SPARQL**.



KG


Grade
Course
School
AlcoholUse
Location
AddressType
boolean
RelationshipStatus
hasEducationLevel
enrolledIn
achievedGrade
wantsHigherEdu
hasInternet
hasAlcoholUse
attends
hasRelationshipStatus
hasSex
livesAt
Parent
hasParent
string
hasJob
EducationLevel
hasHealthStatus
Job
G2
G1
HealthStatus
hasAge
StudyHabit
hasFreeTime
Guardian
integer
G3
hasStudyHabit
hasGuardian
Student
goesOut
hasAbsences
range
domain

KG Embeddings

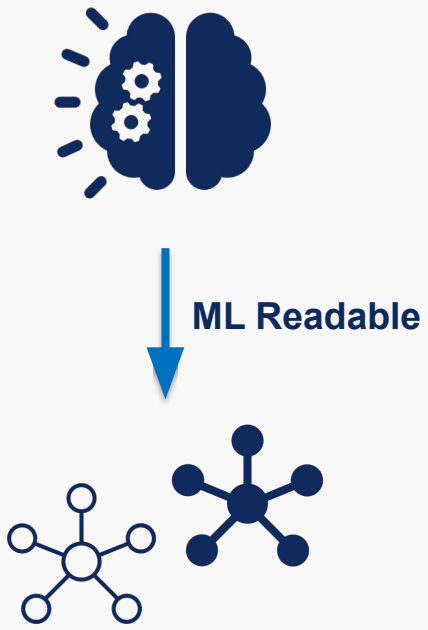
KG EMBEDDINGS

To use the knowledge graph in predictive modeling, we applied **embedding techniques** using PyKEEN. These translate graph structures into fixed-length vector representations that encode relational meaning.

We tested:

- **TransE** – Simple translation-based embedding
- **DistMult** – For symmetric relations
- **ComplEx** – Supports asymmetric relations with complex numbers
- **RotatE** – Models relations as rotations in complex space
- **ConvE** – Uses 2D convolutions to capture intricate patterns

Each embedding model was paired with different regression models.



Prediction Models

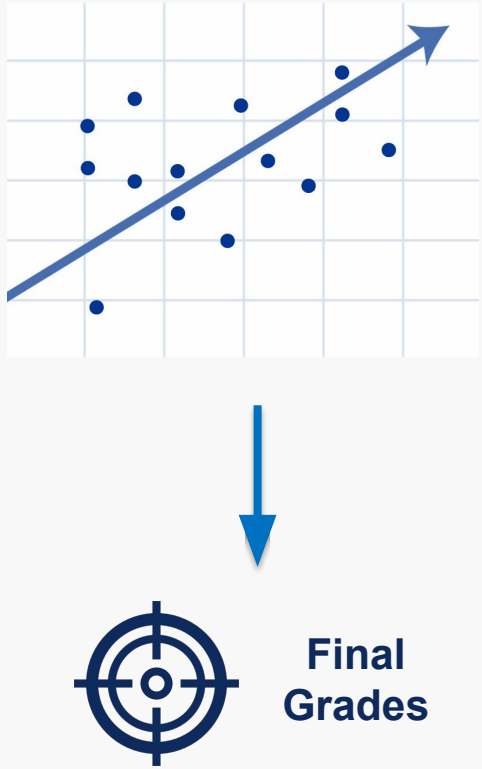
PREDICTION MODELS

Once embeddings were generated, we used them as input features for regression models to **predict students' final grades**.

We experimented with:

- **Linear Regression** – For baseline performance
- **Decision Tree** – Interpretable non-linear model
- **Random Forest** – Ensemble model to reduce overfitting
- **Gradient Boosting** – Highly accurate model correcting prior errors

We evaluated every combination of embedding and regressor to identify the best-performing pipeline.



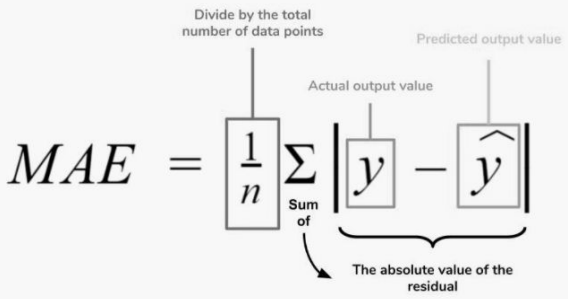
Evaluation

EVALUATION

We evaluated the models using three standard metrics:

- **MAE** (Mean Absolute Error) - Average error magnitude
- **RMSE** (Root Mean Squared Error) - Penalizes large errors
- **R²** (Coefficient of Determination) - Proportion of variance explained

These gave us both statistical and visual confidence in model performance.



$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

Results

RESULTS

Best model was **Random Forest** with **DistMult**:

- **MAE:** 1.14
- **RMSE:** ~1.72
- **R²:** 0.85

This means, on average, our predictions differ from the actual grade by just 1.12 points on a 20-point scale.

Gradient Boosting + **ConvE** was another strong combination.

ConvE and DistMult consistently outperformed more complex models like ComplEx, highlighting the effectiveness of these embeddings in capturing relational patterns.

Regressor	Embedding	MSE	MAE	R^2
Random Forest	RotatE	3.11	1.15	0.846
	ComplEx	3.22	1.20	0.842
	TransE	3.50	1.19	0.828
	DistMult	**3.09**	**1.12**	**0.849**
	ConvE	3.62	1.20	0.821
Linear Regression	RotatE	10.10	2.47	0.500
	ComplEx	9.63	2.33	0.528
	TransE	4.82	1.52	0.767
	DistMult	**4.45**	**1.44**	**0.782**
	ConvE	22.98	3.80	-0.127
Decision Tree	RotatE	6.03	1.34	0.691
	ComplEx	7.16	1.58	0.643
	TransE	6.92	1.44	0.664
	DistMult	**5.12**	**1.24**	**0.748**
	ConvE	6.69	1.36	0.678
Gradient Boosting	RotatE	3.19	1.16	0.843
	ComplEx	3.22	1.22	0.841
	TransE	3.83	1.23	0.813
	DistMult	3.78	1.23	0.815
	ConvE	**3.67**	**1.21**	**0.818**

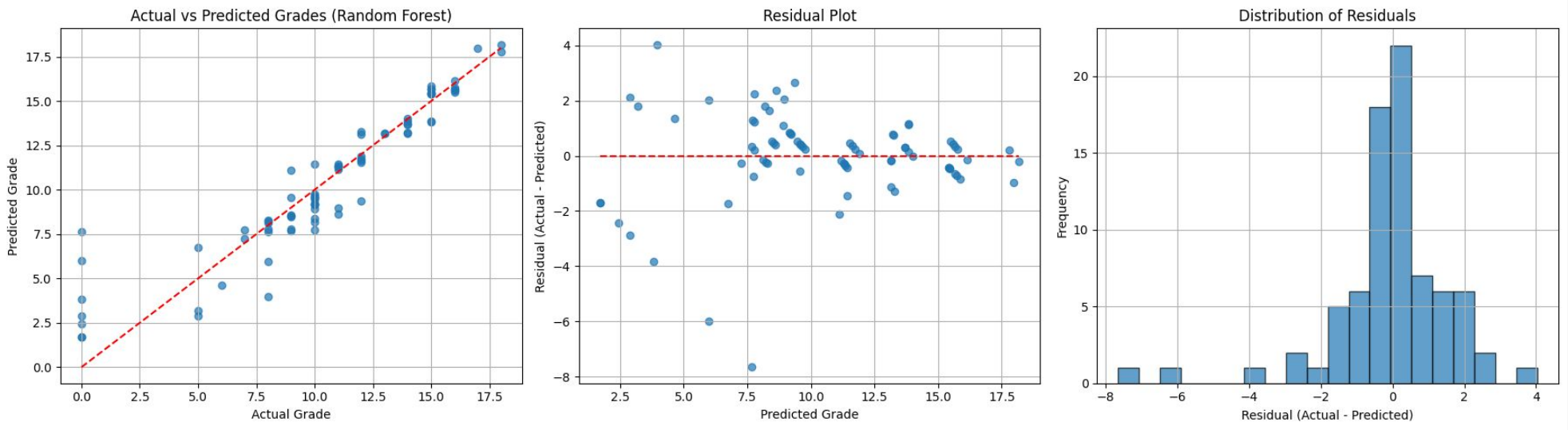
RESULTS

```
Prediction Results (first 20):
Student: http://www.example.org/student-performance#Student_0 | Actual Grade: 14 | Predicted Grade: 15
Student: http://www.example.org/student-performance#Student_100 | Actual Grade: 10 | Predicted Grade: 4
Student: http://www.example.org/student-performance#Student_104 | Actual Grade: 9 | Predicted Grade: 9
Student: http://www.example.org/student-performance#Student_108 | Actual Grade: 0 | Predicted Grade: 5
Student: http://www.example.org/student-performance#Student_111 | Actual Grade: 10 | Predicted Grade: 7
Student: http://www.example.org/student-performance#Student_115 | Actual Grade: 15 | Predicted Grade: 14
Student: http://www.example.org/student-performance#Student_119 | Actual Grade: 0 | Predicted Grade: 2
Student: http://www.example.org/student-performance#Student_122 | Actual Grade: 10 | Predicted Grade: 11
Student: http://www.example.org/student-performance#Student_126 | Actual Grade: 0 | Predicted Grade: 4
Student: http://www.example.org/student-performance#Student_13 | Actual Grade: 10 | Predicted Grade: 8
Student: http://www.example.org/student-performance#Student_133 | Actual Grade: 8 | Predicted Grade: 9
Student: http://www.example.org/student-performance#Student_137 | Actual Grade: 12 | Predicted Grade: 12
Student: http://www.example.org/student-performance#Student_140 | Actual Grade: 13 | Predicted Grade: 13
Student: http://www.example.org/student-performance#Student_144 | Actual Grade: 14 | Predicted Grade: 13
Student: http://www.example.org/student-performance#Student_148 | Actual Grade: 8 | Predicted Grade: 6
Student: http://www.example.org/student-performance#Student_151 | Actual Grade: 11 | Predicted Grade: 11
Student: http://www.example.org/student-performance#Student_155 | Actual Grade: 15 | Predicted Grade: 15
Student: http://www.example.org/student-performance#Student_159 | Actual Grade: 0 | Predicted Grade: 9
Student: http://www.example.org/student-performance#Student_162 | Actual Grade: 9 | Predicted Grade: 8
Student: http://www.example.org/student-performance#Student_166 | Actual Grade: 9 | Predicted Grade: 10
```

RESULTS

Predicted vs. actual grades mostly align along the diagonal, showing **accuracy**.
Residuals are randomly distributed around zero, indicating **no major bias**.
Residual histogram shows **near-normal distribution**, reinforcing the model's **reliability**.

These plots confirm that the embedding + regression approach generalizes well and **avoids overfitting**.



Insights

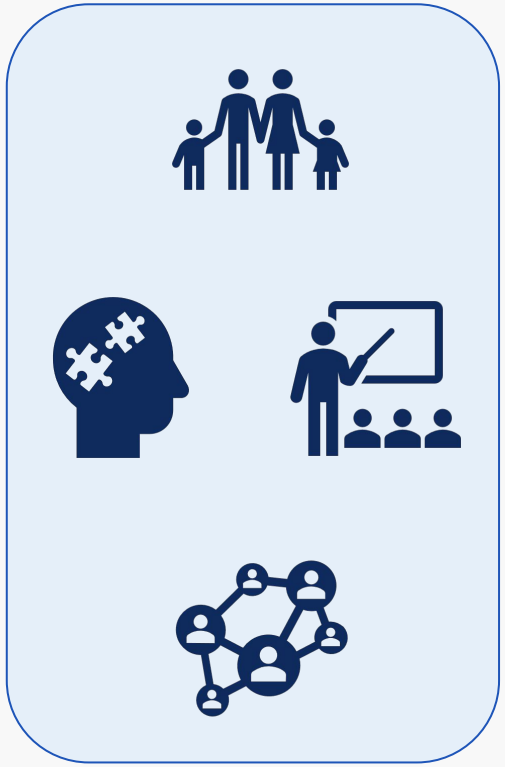
INSIGHTS

Our approach confirms that student performance cannot be explained by isolated features. By semantically modeling students' relationships with their environment, we gain more interpretable, relational, and powerful insights.

Key influencing factors included:

- **Family relationships**
- **Health and behavioral patterns**
- **Early academic performance** (G1 and G2)
- **Social and lifestyle behaviors** (like alcohol use)

The knowledge graph provides both explainability and expandability beyond black-box models.



Limitations

LIMITATIONS

DATASET

Small dataset may not reflect broader populations

GRADES

Some grade ranges were underrepresented (especially failing scores)

DATA

Graph embeddings are sensitive to data sparsity

Future Work

FUTURE WORK

EXTERNAL ONTOLOGIES

01

Add temporal aspects and external ontologies (e.g., Wikidata)

REAL TIME FEATURES

02

Include real-time behavior or learning signals

GNN

03

Try Graph Neural Networks for improved performance

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Thank You